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# Comparison of light absorption levels with different skin phantoms and the Monte Carlo simulation using Fourier-domain optical coherence tomography

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## ABSTRACT

Dermatologic patients have various skin characteristics such as skin tone and pigmentation color. However most studies on laser ablation and treatment only considered laser operating conditions like wavelength, output power and pulse duration. The laser ablation arises from photothermal effect by photon energy absorption. Chromophores like melanin exist as the absorber in the skin. In this study, we painted color to mimic chromophores on in-vivo and in-vitro skin models to demonstrate influence on the laser ablation by skin color. Water-based pens were used to paint color. Cross sectional images of the laser ablation were acquired by Fourier-domain optical coherence tomography (Fd-OCT). Light source to make ablation was a Q-switch diode-pumped Nd:YVO<sub>4</sub> nanosecond laser (532nm central wavelength). Irradiated light energy dose of the laser could not make ablation craters in the control group. However experimental groups showed craters with same irradiation light energy dose. These results show painting on skin increased tissue damage by absorption in painted color without dyeing cells or tissues.

**Keywords:** Laser tissue interaction, Photothermal effect, in-vivo ablation, OCT monitoring, 532nm laser ablation

## 1. INTRODUCTION

The laser ablation was a useful method to remove materials by the photothermal effect [1]. The photothermal effect was generated by absorption of light energy in materials. Vaporization called ablation by the photothermal effect were applied in the medical field as the procedure methods. These procedure methods were used in ophthalmology, cardiology, dentistry, oncology, and dermatology [2-6]. Dermatologic applications were usually used to treat pigmentation [7-8]. It means that the pigmentation region absorbed light energy and get damaged by the photothermal effect. In addition, collateral regions of the photothermal damaged zone get injury by thermal diffusion. It is the major cause of pain and PIH (post inflammation hyperpigmentation and hypopigmentation) [9-10]. Many previous studies modulated laser irradiation conditions including wavelength, output power, and pulse duration to minimize pain as well as ambient damage in a laser focal point by thermal energy [11-18]. Although many researches modulate laser irradiation conditions, influence on the laser tissue interaction by target color had not been reported. Patients who want to treat pigmentation had various skin tone and pigmentation color. The skin tone and pigmentation color vary with chromophore called melanin. It means that the skin of each person had different absorption coefficient of light by difference of skin tone. We investigated correlations between laser tissue interaction and skin color. The laser ablation experiments were performed with painting color on skin to mimic chromophores.

The laser with 532-nm wavelength was used to deliver light energy on the in vitro and in vivo tissue samples. Moreover, Fourier-domain optical coherence tomography (Fd-OCT) was set up to acquire cross-sectional images of laser tissue interactions [19-21]. OCT was non-invasive imaging technology and appropriate tool to observe biological tissue [22]. The laser for ablation and the light source of OCT simultaneously penetrated the same objective lens of the OCT sample arm for ablation monitoring. Acquired OCT cross-sectional images showed craters as the result from the laser ablation.

## 2. METHODS AND MATERIALS

### 2.1 Fourier-domain optical coherence tomography and 532-nm Nd:YVO4 laser system

To observe laser tissue interaction, a Fourier-domain optical coherence tomography (Fd-OCT) and a 532-nm Nd:YVO4 laser were used. A Sample arm of OCT system was designed to combine lasers for ablation and monitoring using the dichroic mirror (FF662-FDi01, Semrock, Rochester, NY, USA) and the objective lens (AC254-030-B-ML, Thorlabs, Newton, NJ, USA). Figure 1 indicated schematic design of OCT and the laser. A broadband light source was the superluminescent diode (SLD, SUPERLUM, Carrigtwohill, Ireland, central wavelength = 860, bandwidth = 135nm, output power = 3mW) with an isolator. The fiber coupler (AC Photonics, Santa Clara, CA, USA) was used for beam delivery and splitting. A reference arm consists of a silver coated reference mirror and a neutral density filter wheel. A sample arm includes the 2-Axis galvo scanner (Cambridge Technology, Bedford, MA USA) for B-scan. The diffraction grating (1200L/mm at 840nm, Wasatch Photonics, Inc. Durham, NC, USA), a focusing lens and the CMOS line scan camera (active pixels: 2048) (spL4096-140km, Basler AG, Ahrensburg, Germany) were built for spectrometer. The high-performance frame grabber (National Instruments, Austin, TX, USA) was installed to acquire fringe patterns in workstation. Image processing software was written in LabVIEW (National Instrument, Austin, TX, USA). The 532-nm laser for ablation was the Q-switch diode-pumped Nd:YVO4 nanosecond laser. Laser specification was 15-ns pulse width at 20-kHz repetition rate and 1.6 Gaussian beam diameter aperture.

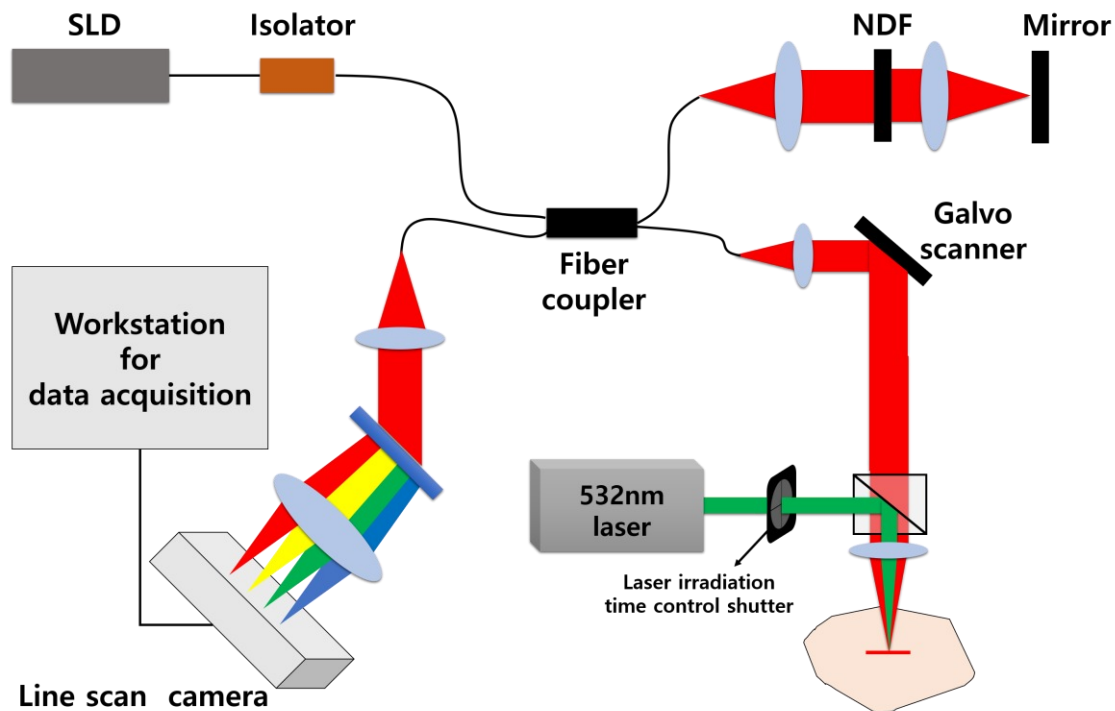


Figure 1. Observation system of the laser ablation consists of Fd-OCT & 532-nm laser. SLD : superluminescent diode, NDF : neutral density filter,

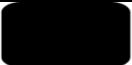
### 2.2 Tissue samples

In vitro porcine skin and in-vivo hairless mice were used to observe the laser ablation. Porcine skin was one of the most popular samples for skin models of percutaneous absorption experiments. All the skin samples were depilated to prevent influence on the hair. Zoletil and rompun solutions were injected into the femoral region of hairless mice to handle it. Ablated regions of samples were located as close as possible for uniform skin properties. Experiments using in vivo hairless mice model were approved by the institutional animal care and use committee (IACUC) of Dankook University.

2.3 Painting on samples

Dyes of water based pens (Plus pen 3000, MONAMI, Yongin, South Korea) were used to mimic chromophores. To make various experimental groups, four colors (Red, Green, Blue, Black), were thinly painted on samples. Water based pens consist of ethylene glycol (CAS No.107-21-1), glycerine(CAS No. 56-81-5), water (CAS No.7732-18-5) and components used as a dye. Before the ablation experiment, chemical effects by each solution (ethylene glycol, glycerine, water, ethylene glycol + glycerine, ethylene glycol + water, glycerine + water, ethylene glycol + glycerine + water) were demonstrated. Differences by chemical effects of each solution did not exist.

Table 1. water based pens

| color                                                                             | Component                                    | Color code | CMYK                                |
|-----------------------------------------------------------------------------------|----------------------------------------------|------------|-------------------------------------|
|  | Ethylenglycol, Glycerine<br>Water, Red dye   | #E60000    | C : 10   M : 100<br>Y : 100   K : 0 |
|  | Ethylenglycol, Glycerine<br>Water, Green dye | #00FF00    | C : 100   M : 0<br>Y : 100   K : 0  |
|  | Ethylenglycol, Glycerine<br>Water, Black dye | #000000    | C : 0   M : 0<br>Y : 0   K : 100    |
|  | Ethylenglycol, Glycerine<br>Water, Blue dye  | #0040FF    | C : 100   M : 75<br>Y : 0   K : 0   |

C : Cyan, M : Magenta, Y : Yellow, K : black

2.4 Laser irradiation time control

The external shutter was built to control precise laser irradiation time. External TTL (transistor-transistor logic) signals were generated by the microcontroller unit (MCU; Arduino Uno). The shutter open time was shorter than input signal as 6ms because shutter open response was delayed by 21ms and shutter close response was delayed 15ms.

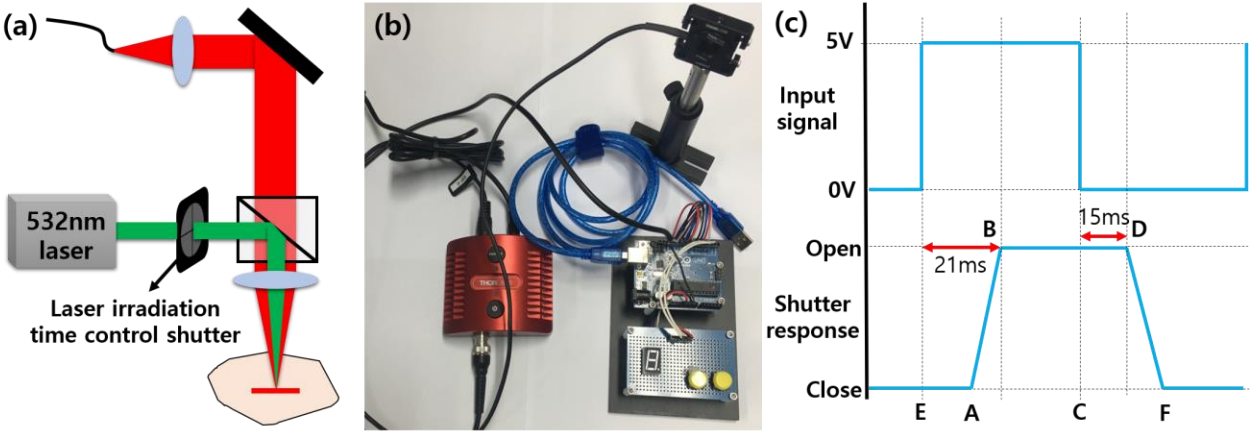


Figure 2. (a) a schematic image of the sample arm, (b) an image of the external shutter, (c) timing diagram of the external shutter

### 3. RESULTS

The cross-sectional images of the laser ablation were acquired during laser irradiation. The images of ablation results were captured and processed after laser exposure for 1 second. The Nd:YOV4 laser was operated in continuous wave mode at 532-nm wavelength. The focused beam of laser by objective lens was 0.8mm beam radius and 1.2 beam propagation factor(M2). To get B-scan images, a 2-axis galvo scanner was used. The scan range was 3.25 mm in all experiments.

#### 3.1 Porcine skin

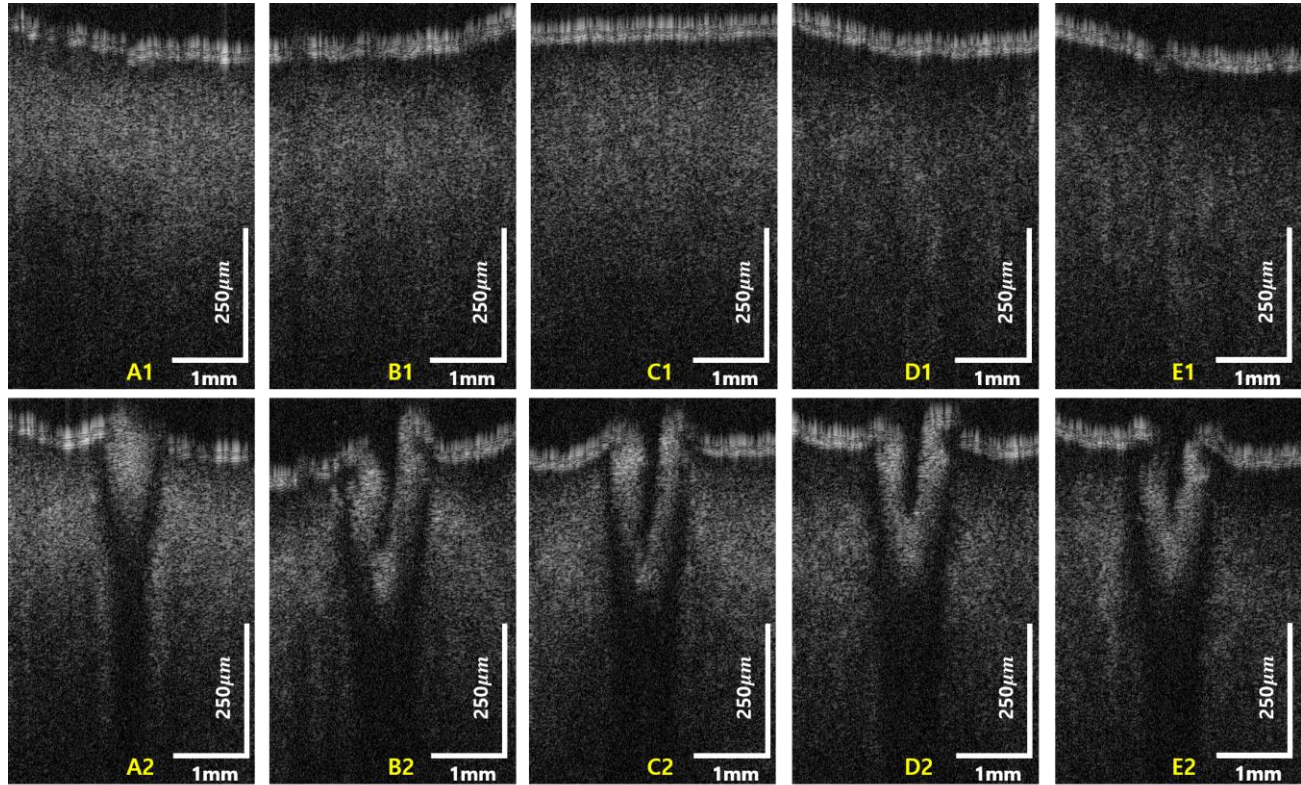


Figure 3. Cross sectional images of the laser ablation for porcine skin painted with four colors. (A1), (B1), (C1), (D1) and (E1) were state before laser irradiation. After laser irradiation, changes and craters were shown in (A2), (B2), (C2), (D2) and (E2). (A1,2): the control group without coloring, (B1,2): an experimental group with painting red, (C1,2): an experimental group with painting green, (D1,2): an experimental group with painting blue, (E1,2): an experimental group with painting black. All samples were irradiated by 525mW power for 1 second (total dose: 0.525J)

In Figure 3, all tissues were damaged by the laser irradiation. In control group, (A2) showed only a change by the laser irradiation. However, all experimental groups (B2), (C2), (D2) and (E2) showed ablated craters. Ablation craters were generated by photothermal effects. Photothermal effects were phenomenon of light absorption. It means that painting color on the skin increased absorption efficiency of the laser.

In Figure 4, All experimental groups (B2), (C2), (D2) and (E2) also showed ablated craters. However, a little change did not exist in the control group (A2). All experimental and control groups of figure 3 and 4 were irradiated same energy dose. However, control groups (A2) of figure 3 and 4 showed different results. It means characteristics of samples like thickness, tone, position of pore could affect the results.

### 3.2 Hairless mouse skin

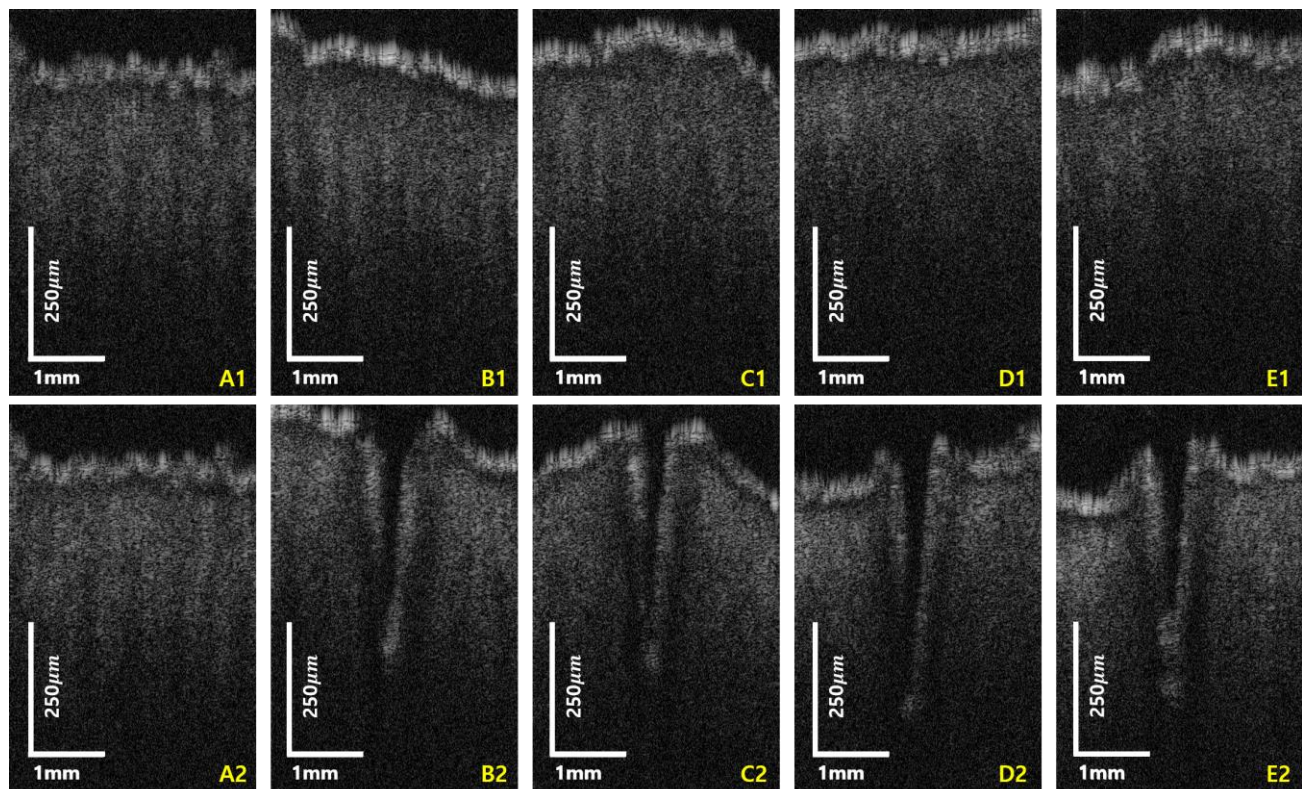


Figure 4. Cross sectional images of the laser ablation for hairless mouse skin painted with four colors. (A1), (B1), (C1), (D1) and (E1) were state before laser irradiation. After laser irradiation, changes and craters were shown in (A2), (B2), (C2), (D2) and (E2). (A1,2): the control group without coloring, (B1,2): an experimental group with painting red, (C1,2): experimental group with painting green, (D1,2): an experimental group with painting blue, (E1,2): an experimental group with painting black. All samples were irradiated by 525mW power for 1 second (total dose: 0.525J)

## 4. CONCLUSIONS

In this research, we demonstrated the effects of skin color on laser ablation by painting color on skin to mimic chromophore. Painting color on skin using water-based pens could increase ablation efficiency without dyeing cells or tissues and injection. These results could be applied to some medical applications. These results mean that it is possible to ablate with low level laser which lack energy to normal skin. In dermatology, painting on skin could help to treat pigmentation with low energy lasers. In addition, it could be used to reduce mistakes as marking target region.

In order to investigate precise experiments, we will try to make homogeneous phantoms which include different quantity of real melanin in the future works. Because all experimental and control groups showed a little difference by different characteristics like skin thickness, tone, pore.

## 5. ACKNOWLEDGEMENTS

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